

# LayerNorm#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.normalization.LayerNorm.html>

## Description:

Applies Layer Normalization over a mini-batch of inputs. This layer implements the operation as described in the paper Layer Normalization. The mean and standard-deviation are calculated over the last  $D$  dimensions, where  $D$  is the dimension of `normalized_shape`. For example, if `normalized_shape` is (3, 5) (a 2-dimensional shape), the mean and standard-deviation are computed over the last 2 dimensions of the input (i.e. `input.mean((-2, -1))`).  $\gamma$  and  $\beta$  are learnable affine transform parameters of `normalized_shape` if `elementwise_affine` is `True`. The variance is calculated via the biased estimator, equivalent to `torch.var(input, unbiased=False)`. Unlike Batch Normalization and Instance Normalization, which applies scalar scale and bias for each entire channel/plane with the affine option, Layer Normalization applies per-element scale and bias with `elementwise_affine`.

## Function Signature

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```
class
torch.nn.modules.normalization.LayerNorm(normalized_shape,
eps=1e-05, elementwise_affine=True, bias=True, device=None,
dtype=None)[source]#
```

Parameters

Variables

Shape:

## Code Examples

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```
>>> # NLP Example
>>> batch, sentence_length, embedding_dim = 20, 5, 10
>>> embedding = torch.randn(batch, sentence_length,
embedding_dim)
>>> layer_norm = nn.LayerNorm(embedding_dim)
>>> # Activate module
>>> layer_norm(embedding)
>>>
>>> # Image Example
>>> N, C, H, W = 20, 5, 10, 10
>>> input = torch.randn(N, C, H, W)
>>> # Normalize over the last three dimensions (i.e. the channel
and spatial dimensions)
>>> # as shown in the image below
>>> layer_norm = nn.LayerNorm([C, H, W])
>>> output = layer_norm(input)
```

## Notes & Warnings

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### NOTE

Unlike Batch Normalization and Instance Normalization, which applies scalar scale and bias for each entire channel/plane with the affine option, Layer Normalization applies per-element scale and bias with `elementwise_affine`.

## Detailed Documentation

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### Documentation

Applies Layer Normalization over a mini-batch of inputs.

This layer implements the operation as described in the paper [Layer Normalization](#)

The mean and standard-deviation are calculated over the last  $D$  dimensions, where  $D$  is the dimension of `normalized_shape`. For example, if `normalized_shape` is `(3, 5)` (a 2-dimensional shape), the mean and standard-deviation are computed over the last 2 dimensions of the input (i.e. `input.mean((-2, -1))`).  $\gamma$  and  $\beta$  are learnable affine transform parameters of `normalized_shape` if `elementwise_affine` is `True`. The variance is calculated via the biased estimator, equivalent to `torch.var(input, unbiased=False)`.

Unlike Batch Normalization and Instance Normalization, which applies scalar scale and bias for each entire channel/plane with the affine option, Layer Normalization applies

per-element scale and bias with `elementwise_affine`.

This layer uses statistics computed from input data in both training and evaluation modes.

`normalized_shape` (int or list or `torch.Size`) –

input shape from an expected input of size

If a single integer is used, it is treated as a singleton list, and this module will normalize over the last dimension which is expected to be of that specific size.

`eps` (float) – a value added to the denominator for numerical stability. Default:  $1e-5$

`elementwise_affine` (bool) – a boolean value that when set to `True`, this module has learnable per-element affine parameters initialized to ones (for weights) and zeros (for biases). Default: `True`.

`bias` (bool) – If set to `False`, the layer will not learn an additive bias (only relevant if `elementwise_affine` is `True`). Default: `True`.

`weight` – the learnable weights of the module of shape `normalized_shape` when `elementwise_affine` is set to `True`. The values are initialized to 1.

`bias` – the learnable bias of the module of shape `normalized_shape` when `elementwise_affine` is set to `True`. The values are initialized to 0.

Input:  $(N, *, N, *)$

Output:  $(N, *, N, *)$  (same shape as input)

